

Prasun Kumar

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Technical Lead | AI/ML Architect | Generative AI Solution Architect
Technical Lead ★ AI/ML Architect ★ GenAI Solution Architect

Seeking a challenging role as a Technical Lead in AI&ML/Architect, to leverage extensive experience and expertise of 8.5+ years to drive innovative projects, lead cross-functional teams, and contribute to the development of cutting-edge solutions in artificial intelligence and machine learning. The goal is to lead impactful initiatives, foster collaboration, and deliver high-quality AI and ML solutions that drive business growth and technological advancement.

Professional Summary

- Technical Lead and AI/ML Architect with over 8.5 years of experience designing and delivering enterprise-scale AI, Machine Learning, Generative AI, Intelligent Automation, and Cloud-Native solutions.
- Proven expertise in Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Azure OpenAI, Vector Search, AI Agents, Intelligent Document Processing, MLOps, and enterprise system integrations.
- Experienced in leading cross-functional teams, defining solution architecture, engaging with clients and stakeholders, and delivering production-grade AI platforms on Azure and AWS.
- Strong background in Generative AI, NLP, Computer Vision, Deep Learning, Automation Engineering, and Enterprise Integrations across Healthcare, Insurance, HRTech, EdTech, Background Verification, and Learning Platforms.
- Passionate about building scalable, secure, and responsible AI systems that drive measurable business outcomes.

Core Skills

• GPT-4 • Azure OpenAI • Llama • Gemma • Qwen • Prompt Engineering • Retrieval-Augmented Generation (RAG) • Agentic AI • Vector Databases • AI Governance • Deep Learning	• Microsoft Azure • AWS • Azure AI Search • Azure Functions • Azure Service Bus • Azure Blob Storage • Event-Driven Architecture • Microservices • Serverless Computing • LoRA • QLoRA • Computer Vision	• MLflow • Databricks • Model Monitoring • CI/CD • Docker • Kubernetes • TensorFlow • PyTorch • Natural Language Processing (NLP)
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Experience		
Company	Role	Duration
Harbinger Group	Technical Lead – AI & ML	Apr 2025 – Present
Harbinger Group	Senior Software Engineer	April 2021-Mar 2025
Extentia Information Technology	Software Engineer	Nov 2019 – Apr 2021
Symantec Software India Pvt Ltd	Associate IT Applications Specialist	Jul 2017 – Oct 2019
Symantec Software India Pvt Ltd	Master Level Intern	Jan 2017 – Jul 2017

Key Projects

GenAI-Powered Email-to-Case Automation Platform

Role: Technical Lead / AI Solution Architect

- Architected and led the development of a GenAI-powered email-to-case automation platform processing inbound emails and unstructured documents into structured case records.
- Designed an event-driven Azure architecture leveraging Azure Functions, Service Bus, Blob Storage, Azure Database for MySQL, and Azure Monitor.
- Implemented intelligent document processing using Azure AI Document Intelligence and GPT-4 for extracting business-critical information from emails, PDFs, images, and scanned documents.
- Integrated Microsoft Graph API for automated email ingestion and notification workflows.
- Designed enterprise-grade resiliency using retry mechanisms, dead-letter queues, idempotent processing, monitoring, and alerting.
- Built secure integrations with TrackOps APIs for automated case creation and workflow orchestration.
- Established AI governance controls including confidence scoring, auditability, validation workflows, and human-in-the-loop review.
- Achieved over 60% improvement in extraction accuracy and significantly reduced manual case processing effort.

Technologies: Azure OpenAI, GPT-4, Azure AI Document Intelligence, Azure Functions, Service Bus, Blob Storage, MySQL, Microsoft Graph API, Python, Azure Monitor

Requirements Discovery Agent – Enterprise Knowledge Discovery Platform

Role: Technical Lead – Generative AI & Knowledge Discovery Platform

- Architected an enterprise Retrieval-Augmented Generation (RAG) platform enabling intelligent search and conversational access to organizational knowledge.
- Designed automated document ingestion pipelines using Azure Blob Storage, Azure Document Intelligence, Azure AI Search, and Vector Search.
- Developed chunking, embedding generation, metadata enrichment, semantic indexing, and retrieval workflows.
- Implemented hybrid retrieval using vector similarity, keyword search, and metadata filtering.
- Defined vector schemas, retrieval quality frameworks, chunk optimization strategies, and evaluation methodologies.
- Collaborated with engineering teams to establish scalable cloud-native AI architecture and API standards.
- Conducted retrieval benchmarking and optimization to improve answer relevance and reduce hallucinations.

Technologies: Azure AI Search, Azure Document Intelligence, Vector Search, RAG, Large Language Models, Python, Azure Storage

Behavioral Intelligence Platform – LLM Fine-Tuning & Multi-Adapter

Architecture

Role: Technical Lead

- Led training, evaluation, and deployment of multiple behavior scoring models using Llama, Gemma, and Qwen architectures.
- Implemented QLoRA fine-tuning, adapter merging, and dynamic multi-adapter serving strategies.
- Built Databricks-based experimentation, evaluation, and monitoring dashboards.
- Conducted benchmarking across multiple LLM architectures and multilingual datasets including Japanese and English.
- Optimized storage utilization by implementing cleanup and lifecycle management processes saving over 77 TB of storage.

Technologies: Llama, Gemma, Qwen, QLoRA, Databricks, Python

AI-Powered Intelligent Learning Content Generator

Role: Technical Lead – Generative AI for Learning

- Developed a generative AI platform that automatically creates instructional content, quizzes, and assessments tailored to specific learning objectives and audiences.
- Designed retrieval-augmented generation pipelines that integrate domain knowledge bases to ensure accurate and contextually relevant content.
- Implemented content quality evaluation workflows, including human-in-the-loop review, to maintain pedagogical standards and reduce hallucination.
- Achieved substantial reduction in course development time and enabled scalable personalized learning content delivery.

Technologies: GPT-4, RAG, Azure AI Search, Python, Cloud Functions

Virtual HR Assistant & Candidate Screening Agent

Role: Technical Lead – Conversational AI & NLP

- Designed and implemented a conversational AI agent that handles employee HR inquiries, automates routine tasks, and screens job candidates based on predefined criteria.
- Integrated large language models with enterprise HR systems to provide real-time responses and automate data retrieval and updates.
- Developed candidate screening workflows leveraging NLP and LLMs to extract key information from resumes and assessments.
- Enhanced HR service efficiency and improved candidate experience by reducing response times and ensuring consistent information delivery.

Technologies: GPT-4, RAG, Azure Bot Service, Python, HRIS Integrations

Multi-Agent Workflow Automation Platform

Role: Technical Lead – AI Agents & Orchestration

- Developed a multi-agent AI orchestration platform that automates complex enterprise workflows across HR, CRM, and support systems.
- Designed agentic workflows enabling dynamic tool selection, memory retrieval, reasoning, and decision-making using large language models.
- Implemented custom tool integrations to interface with various enterprise systems, enabling automated data updates and task completion.
- Improved operational efficiency by reducing manual intervention and enabling end-to-end process automation.

Technologies: GPT-4, LangChain, Azure Functions, Service Bus, Python

Technical Leadership

- Led cross-functional teams of AI engineers, software engineers, QA engineers, and business stakeholders.
- Owned end-to-end solution architecture, technical design reviews, and delivery planning.
- Conducted client-facing architecture discussions and requirement workshops.
- Mentored junior engineers on AI/ML development, cloud architecture, and engineering best practices.
- Defined coding standards, review processes, and deployment strategies.